

SUPPLY CHAIN EXCEPTION CONTROL SYSTEM

Frozen Operational Exception Lifecycle and Operating Model

Version 1.0 | 20 July 2026

CONTROL POSITION Candidate Risk is an analytical assessment. The operational lifecycle begins at Open. Escalation, SLA breach and reopening are events or conditions, not statuses. Resolved and Closed are human-controlled in the MVP; automation may detect evidence but may not silently resolve or close a case.

Document field	Definition
Document status	Frozen portfolio design baseline; proposed assumptions, not a validated company policy
Prepared for	Portfolio project - simulated Australian mid-sized distributor/manufacturer
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Governing sources	Project Charter v1.0 and Business Requirements Specification v1.0, both dated 19 July 2026
Primary operational grain	One purchase-order-line risk episode
Operational time zone	Australia/Melbourne
Scope boundary	Operating model only; no physical data model, code, dashboard, notification, deployment or AI implementation

EVIDENCE LIMITATION This design has not been operationally validated. All SLAs, monitoring periods, authority thresholds and calendar rules are proposed portfolio assumptions to be demonstrated with synthetic scenarios.

Document control and reading guide

This operating model resolves the pre-modelling decisions identified in BRS Appendix E (DL-002, DL-004 and DL-011 to DL-015). Where it intentionally changes a preliminary BRS rule, the change is identified in Section 12 and the BRS revision list. The charter remains authoritative for the business problem, objectives, scope, control principles and portfolio limitations.

Authority order	Use in this document
1. Approved Charter v1.0	Governing business problem, objectives, scope, success criteria and limitations.
2. BRS v1.0	Governing requirements baseline and traceability identifiers; provisional lifecycle rules are refined here.
3. This frozen operating model	Approved semantic interpretation for lifecycle, episode, ownership, SLA, approval, resolution and human/automation boundaries before logical modelling.
4. Later technical design	Must implement these semantics; it may not redefine them implicitly.

INTERPRETATION RULE A candidate score, a low current score, an escalation flag or a recommendation is never evidence that an exception has been resolved or closed.

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1. Operational process overview

1.1 End-to-end control flow

Step	Stage	Control activity	System role	Human control
1	Source data arrival	Receive dated PO, receipt, inventory, demand, confirmation, supplier, item and site inputs; create immutable run identity and Melbourne timestamps.	Automated	Administrator monitors run; no operational case action.
2	Data validation	Validate keys, references, quantity/date logic, units, completeness, freshness and control totals. Stop critical failures; quarantine permitted non-critical defects.	Automated gate	Administrator investigates defects and authorises corrected reprocessing; cannot rewrite prior run history.
3	Candidate risk assessment	Score each eligible PO line/run; retain version, contributions and disposition. Candidate Risk remains analytical and is not shown as an operational status.	Automated	Rule owners approve versions outside the daily workflow.
4	Exception creation	If the opening rule is met, no active duplicate exists and no applicable suppression control blocks creation, create one Open risk episode.	Automated	Humans cannot delete an automatically created episode; they may challenge it through controlled evidence routes.

Step	Stage	Control activity	System role	Human control
5	Assignment	Resolve the owner using the frozen hierarchy. A valid mapping moves Open to Assigned; unresolved mapping remains Open in the visible unassigned queue.	Automated first; human fallback	Manager or authorised analyst assigns/reassigns with reason. Critical reassignment is controlled.
6	Investigation	Owner explicitly accepts work, reviews risk reasons and exposure, checks source validity and records cause/contacts/evidence.	Human-controlled	Automation may refresh facts and score snapshots only.
7	Action agreed	Record the selected action, accountable actor, next action/commitment, due time, evidence and any required approval.	Human-controlled	A system suggestion has no effect until a human records the plan.
8	Monitoring	Confirm action initiation and define the objective event/date to monitor. Track commitments, receipts, transfers, substitutions, demand changes and source corrections.	Human entry; automated monitoring	The system creates reminders, breach and possible-resolution-evidence events but does not resolve.
9	Resolution	Authorised human selects a structured category, records evidence, residual risk and approvals, and confirms that risk is addressed, removed or formally accepted.	Human-controlled	System validates fields and may recommend eligibility; no silent transition.
10	Closure	Authorised reviewer verifies the resolution evidence and monitoring condition, records closure rationale, and formally completes the episode.	Human-controlled	Closed history is not reopened by editing; recurrence creates a controlled reopen event and, after closure, a linked successor episode.
11	Suppression or reopening	Suppression temporarily removes an active episode from normal work under reason, authority and expiry. Expiry/revocation reactivates it. Material recurrence after closure opens a linked successor.	System expiry/recurrence detection; human approval/review	No deletion, indefinite suppression or reset of prior SLA/audit history.

1.2 Automated versus human-controlled boundary

Automation may	Automation may not
Ingest, validate, quarantine, score and explain candidate evaluations.	Approve a material override or accept residual risk.
Create Open when a qualifying candidate has no active duplicate.	Move an exception into Resolved or Closed in the MVP.
Assign using an approved deterministic mapping and expose failures.	Conceal an unassigned, breached or expired-suppression condition.
Refresh risk facts, detect escalation/SLA breach and create immutable events.	Reset an SLA because ownership or severity changed.
Detect receipt, substitution, cancellation or correction evidence and propose a resolution route.	Treat a falling score or supplier promise alone as conclusive resolution.
Expire suppression and reactivate the episode; detect material post-closure recurrence.	Delete, rewrite or backdate lifecycle, action, approval or evidence history.

2. Final lifecycle

RECOMMENDED FINAL LIFECYCLE Open -> Assigned -> Investigating -> Action Agreed -> Monitoring -> Resolved -> Closed, with controlled shortcut resolution from Investigating or Action Agreed when the risk is objectively removed. Suppressed is a temporary dormant state reachable only from an active state. Reopened is an event, not a state.

Open

Lifecycle attribute	Frozen definition
Business meaning	A qualifying candidate has become an operational control record, but no accountable individual is yet recorded.
Entry criteria	Successful validated run; opening rule met; no active duplicate; no applicable candidate suppression; episode identity assigned.
Mandatory fields	Episode/reference; opening candidate and run; site; current severity and reasons; opened-at; source freshness; assignment and action-plan due times; prior linked episode if any.
Permitted actions	Assign; add triage note; request suppression; link a correction; escalate unassigned condition.
Exit criteria	Valid owner recorded -> Assigned; approved suppression -> Suppressed.
Authorised roles	System creates. Procurement analyst or manager may assign. Manager approves suppression.
SLA behaviour	Assignment and action-plan clocks start at Open; after-hours time begins consuming at next business opening.
Entry control	Automatic

Assigned

Lifecycle attribute	Frozen definition
Business meaning	One accountable procurement owner is recorded; ownership alone is not evidence that investigation has begun.
Entry criteria	Approved automatic mapping or authorised manual assignment; owner active and eligible; assignment event retained.
Mandatory fields	Owner; ownership basis/mapping version; assigned-at; assigning actor/system; manual reason if applicable; temporary-delegation reference where used.
Permitted actions	Accept work; reassign; delegate; add note/evidence; request suppression; system refreshes risk.
Exit criteria	Owner explicitly begins work -> Investigating; approved suppression -> Suppressed.
Authorised roles	Assigned owner, procurement manager; system for mapped assignment.
SLA behaviour	Assignment clock stops. Action-plan clock continues; reassignment does not reset either clock.
Entry control	Automatic or human-controlled

Investigating

Lifecycle attribute	Frozen definition
Business meaning	The accountable owner has acknowledged the exception and is determining validity, cause, consequence and response.
Entry criteria	Owner or authorised manager explicitly starts investigation; current facts and score explanation are visible.
Mandatory fields	Investigation start; investigator; initial validity assessment; cause/status; contact record where relevant; next review; evidence references; current residual exposure.
Permitted actions	Record supplier contact; request planner assessment; correct source-data link; propose action/approval; resolve through rapid-resolution route; reassign; request suppression.
Exit criteria	Complete plan -> Action Agreed; objective/judgement resolution with all evidence -> Resolved; approved suppression -> Suppressed.
Authorised roles	Procurement analyst; inventory planner contributes; manager may take over.

Lifecycle attribute	Frozen definition
SLA behaviour	Action-plan clock continues. Supplier waiting does not pause it. Approved data/system hold may pause.
Entry control	Human-controlled

Action Agreed

Lifecycle attribute	Frozen definition
Business meaning	A documented and authorised plan exists, but implementation or the monitored commitment is not yet confirmed as initiated.
Entry criteria	Action type, accountable actor, next-action date, monitoring condition, evidence and all approvals are complete.
Mandatory fields	Action category; rationale; action owner; agreed-at; target/commitment; next action; evidence; approval reference; residual risk; expected monitoring event/date.
Permitted actions	Initiate action; amend plan by appended event; seek approval; return for investigation if invalid; resolve if action completes immediately; request suppression.
Exit criteria	Action initiated -> Monitoring; plan withdrawn/new facts -> Investigating; immediate verified outcome -> Resolved; approved suppression -> Suppressed.
Authorised roles	Procurement analyst; planner for inventory actions; manager for controlled actions.
SLA behaviour	Action-plan SLA stops at first compliant Action Agreed entry. Later plan failure creates monitoring breach; it does not erase original response performance.
Entry control	Human-controlled

Monitoring

Lifecycle attribute	Frozen definition
Business meaning	An agreed response or supplier commitment is in progress and a defined outcome/date is being watched.
Entry criteria	Initiation confirmed; monitoring condition and deadline recorded; owner remains accountable.
Mandatory fields	Monitoring start; monitored condition; expected event/date; next review; owner; evidence; reminder/escalation schedule; residual risk; current score snapshots.
Permitted actions	Record checkpoint; change next review; escalate; investigate breach; submit resolution; reassign; request suppression.
Exit criteria	Risk addressed/removed/accepted -> Resolved; commitment breach/new cause -> Investigating; approved suppression -> Suppressed.
Authorised roles	Procurement analyst; inventory planner for stock actions; manager oversees critical/breached cases.
SLA behaviour	Response clock already stopped. Commitment/next-action timers run; waiting on supplier does not pause monitoring. Breaches create events, not states.
Entry control	Human entry; automated surveillance

Resolved

Lifecycle attribute	Frozen definition
Business meaning	An authorised human has concluded that the operational risk was addressed, objectively removed or formally accepted with evidence. Work remains subject to closure review.

Lifecycle attribute	Frozen definition
Entry criteria	Structured resolution category; cause; evidence; resolver; residual risk; monitoring result; approvals; no unresolved mandatory action.
Mandatory fields	Resolution code/time; resolver; cause; evidence; outcome quantity/date; residual-risk statement; approval; proposed closure review date; current rule/score reference.
Permitted actions	Review evidence; add monitoring result; withdraw resolution if evidence invalidates; close when criteria met.
Exit criteria	Authorised review -> Closed; invalid/reversed evidence before closure -> Investigating via Reopened/Resolution Withdrawn event.
Authorised roles	Procurement analyst or planner within category authority; manager for acceptance/override routes. System cannot enter.
SLA behaviour	All response clocks stop. Closure-review expectation begins. If withdrawn before closure, prior SLA does not reset.
Entry control	Human-controlled only

Closed

Lifecycle attribute	Frozen definition
Business meaning	An authorised reviewer has verified resolution and formally completed the risk episode. It is historical, not editable operational work.
Entry criteria	Resolved; required monitoring period complete; evidence resolvable; approvals valid; reviewer authorised; closure reason recorded.
Mandatory fields	Closed-at; reviewer; closure rationale; evidence/approval references; final residual risk; outcome; link to resolution; SLA result; limitations/exceptional-closure flag.
Permitted actions	View/export history; add non-destructive audit annotation. A new material recurrence may create a linked successor episode.
Exit criteria	No state exit. Post-closure recurrence creates a new Open successor and linked Reopened event; it does not change the closed episode status.
Authorised roles	Supply-chain manager or designated closure reviewer; governance concurrence for exceptional closure.
SLA behaviour	Clocks remain stopped. Successor episode starts new clocks; prior SLA remains unchanged.
Entry control	Human-controlled only

Suppressed

Lifecycle attribute	Frozen definition
Business meaning	An active episode is temporarily removed from normal action presentation because an authorised, time-bounded suppression is in force. It is dormant, not resolved or closed.
Entry criteria	Active state Open through Monitoring; reason code and narrative; authorised requester and approver; evidence; start and expiry; scope; no self-approval for material suppression.
Mandatory fields	Prior active state/owner; reason; requested/approved by; start/expiry; evidence; conditions; review date; score history; SLA consumed/remaining; recurrence criteria.
Permitted actions	Review; revoke early; extend only through a new approval; view candidate evaluations; reactivate on expiry/revocation/material threshold.
Exit criteria	Expiry/revocation -> Open if no valid owner, otherwise Assigned; urgent material deterioration may reactivate early. Human then investigates/resolves normally.

Lifecycle attribute	Frozen definition
Authorised roles	Manager approval; governance concurrence for critical extended/repeated suppression; system executes expiry/reactivation.
SLA behaviour	Assignment/action clocks pause only while approved suppression is effective; elapsed consumption is preserved. On reactivation they resume, not reset; an already breached clock stays breached.
Entry control	Human-approved entry; automatic exit

2.1 Non-state control concepts

Concept	Frozen treatment	Reason
Candidate Risk	Analytical record per eligible PO line per successful run, with disposition: below threshold, linked to active episode, blocked by suppression, or opened new episode.	Avoids mixing scoring population with operational workload.
SLA breach	Condition plus immutable event; does not change lifecycle state.	An owner must still see the actual work stage.
Escalation	Event and additional accountability relationship; original owner remains responsible unless explicitly reassigned.	Prevents escalation from becoming an ownership loophole.
Reopened	Event. Before closure, it returns Resolved or Suppressed work to an active state. After closure, it links a new successor episode to the closed predecessor.	Preserves closure metrics and history while avoiding duplicate families.
Possible resolution evidence	System-detected event/flag awaiting human assessment.	Objective signals can be corrected or reversed.

3. Complete state-transition matrix

All permitted transitions are listed below. Any from/to pair not listed is prohibited. Owner, priority, escalation and evidence changes that do not alter the work stage are appended as events and retain the current lifecycle state.

Current	Next	Trigger	Mode	Authority	Reason	Evidence	Approval	SLA effect	Prohibited example
No episode	Open	Qualifying candidate; no active duplicate; not blocked by valid suppression	Automatic	System under approved rule	Opening rule/version	Candidate, score contribution and run refs	None	Starts assignment and action clocks	Candidate directly to Assigned; creating second active episode
Open	Assigned	Valid mapping or authorised manual assignment	Automatic or human	System; analyst; manager	Manual only	Mapping/delegation ref	Critical manual reassignment control if replacing a proposed owner	Stops assignment clock; action clock continues	Open directly to Investigating/Closed
Open	Suppressed	Approved temporary suppression	Human decision	Distinct manager; governance if critical extended/repeated	Mandatory	Basis and expiry	Manager; no self-approval	Pauses clocks; preserves consumption	Suppression without expiry
Assigned	Investigating	Owner explicitly accepts/begins work	Human	Owner or manager	Not normally	Optional initial note	None	Action clock continues	System starts investigation silently
Assigned	Suppressed	Approved temporary suppression	Human decision	Distinct manager	Mandatory	Basis and expiry	Manager	Pauses clocks	Assigned directly to Closed
Investigating	Action Agreed	Complete plan and approvals	Human	Owner; manager for controlled plan	Mandatory	Plan, due date, commitment and approvals	Per approval matrix	Stops action-plan clock	Plan missing next action/evidence
Investigating	Resolved	Rapid objective removal, false positive, cancellation or source correction	Human	Authorised resolver	Mandatory	Category-specific evidence	Per resolution matrix	Stops response clocks; starts closure review	Low score alone; system-only resolution
Investigating	Suppressed	Approved temporary suppression	Human decision	Distinct manager	Mandatory	Basis, evidence, expiry	Manager	Pauses clocks	Owner self-approves material suppression
Action Agreed	Monitoring	Action initiated and monitoring condition defined	Human	Action owner	Not normally	Initiation evidence; expected event/date	Controlled action approval must already exist	Response SLA remains stopped; monitoring timer starts	Monitoring without measurable condition
Action Agreed	Investigating	Plan withdrawn, rejected or invalidated	Human	Owner or manager	Mandatory	Changed facts/failed approval	None	Original SLA result not erased; new next-action timer	Delete or overwrite prior plan
Action Agreed	Resolved	Action completed immediately with evidence	Human	Authorised resolver	Mandatory	Category evidence and residual risk	Per resolution matrix	Stops clocks	Automatic resolution after recommendation
Action Agreed	Suppressed	Approved temporary suppression	Human decision	Distinct manager	Mandatory	Basis/evidence/expiry	Manager	Pauses applicable timers	Indefinite suppression

Current	Next	Trigger	Mode	Authority	Reason	Evidence	Approval	SLA effect	Prohibited example
Monitoring	Resolved	Monitored outcome achieved or residual risk accepted	Human	Authorised resolver	Mandatory	Category evidence and monitoring result	Per resolution matrix	Stops monitoring timer; starts closure review	Receipt flag silently resolves
Monitoring	Investigating	Commitment breached, evidence reversed, action failed or new cause	Automatic trigger + human state change, or human	Owner/manager	Mandatory	Breach/reversal/new-risk evidence	None	Breach remains; no reset	Create duplicate exception for same active episode
Monitoring	Suppressed	Approved temporary suppression	Human decision	Distinct manager	Mandatory	Basis/evidence/expiry	Manager	Pauses timers	Suppress solely because SLA breached
Resolved	Closed	Monitoring period met and evidence verified	Human	Closure reviewer	Mandatory	Resolution, monitoring and approval refs	Manager; governance for exceptional closure	Clocks stay stopped	Closure without resolution evidence; self-review where prohibited
Resolved	Investigating	Resolution evidence invalidated before closure	Human after system alert or human detection	Resolver/manager	Mandatory	Reversal/correction evidence	Manager if prior accepted residual risk	Resume preserved clock/breach; no reset	Rewrite Resolved event
Suppressed	Open	Expiry/revocation with no valid eligible owner	Automatic	System	Expiry/revocation ref	Suppression record and latest candidate	None	Resumes remaining clocks; breach preserved	Remain suppressed past expiry
Suppressed	Assigned	Expiry/revocation with valid eligible owner	Automatic	System	Expiry/revocation ref	Suppression record and owner mapping	None	Resumes remaining clocks; breach preserved	Reset SLA at reactivation
Closed	Open (new linked episode)	Material recurrence/new deterioration after formal closure	Automatic opening; linked Reopened event	System under approved rule	Recurrence rule/version	New candidate plus predecessor link	None	New successor clocks start; predecessor unchanged	Change predecessor from Closed to Open; unlinked duplicate
Any active state	Same state	Reassignment, priority change, escalation, reminder, evidence addition or score refresh	Automatic or human event	Per action authority	Mandatory where human/control change	Before/after and source/evidence	Per approval matrix	No reset; escalation accountability added	Treat event as a fake lifecycle status

3.1 Binding negative controls

ID	Attempt	Required control response
NC-01	Open -> Closed or Assigned -> Closed	Blocked. A case must be investigated, resolved with evidence and then reviewed.
NC-02	Any active state -> Closed	Blocked. Only Resolved -> Closed is valid.
NC-03	Self-approval of material override	Blocked. Requester and approver must be distinct simulated identities; Critical exceptional decisions may require two approvers.

ID	Attempt	Required control response
NC-04	Suppression without expiry, evidence or authorised reason	Blocked. Expiry is mandatory and extension is a new approval, not an edit.
NC-05	Closure without resolution evidence	Blocked unless exceptional-closure route has manager plus governance approval and explicit limitation flag.
NC-06	Automated Resolved or Closed	Blocked in MVP even when an objective receipt event exists.
NC-07	Reassignment resets SLA	Blocked. SLA history and consumed time remain.
NC-08	Priority reduction overwrites calculated score	Blocked. Original score remains visible; approved operational override is separate and time-bound.
NC-09	Delete/rewrite historical action, approval, receipt or transition	Blocked through normal workflow. Corrections append compensating events and retain before/after.
NC-10	Second active episode for same logical PO line	Blocked. Candidate evaluation links to the existing episode.
NC-11	Closed recurrence reuses the closed episode as if closure never occurred	Blocked. Create linked successor episode and retain predecessor outcome/SLA.
NC-12	Supplier waiting pauses SLA	Blocked. Waiting is an operational condition; only approved data/system hold or effective suppression pauses.

3.2 Controlled events that do not change lifecycle state

Event	Required treatment
Risk recalculated	Append score, severity, contributions, rule version and run reference; current state remains.
Owner reassigned/delegated	Append prior/new owner, actor, reason, handover and approval where required; state and SLA remain.
Priority/criticality overridden	Append original/requested/approved value, authority and expiry; calculated score remains unchanged.
Reminder/SLA breach	Append due time, elapsed business minutes and recipient/accountability; do not create Escalated or Breached status.
Evidence added/corrected	Append new evidence or compensating correction linked to the original; never replace prior reference silently.
Approval requested/decided	Append request and decision; lifecycle changes only when the separately authorised action is executed.
Possible resolution detected	Append objective source event and proposed category; human review is required before Resolved.
Audit annotation	Governance reviewer may append an observation; the operational history and state remain unchanged.

4. Exception episode and deduplication policy

4.1 Frozen episode definition

EPISODE DEFINITION One purchase-order-line risk episode is the continuous period of operational attention arising from a materially related delay and/or stockout exposure on one stable source PO line at one receiving site, beginning when the first qualifying candidate opens an exception and ending only when that exception is Closed. Score runs, partial receipts, amendments, changing causes and mitigation actions are observations within the episode unless the PO line identity or accountable operational obligation changes.

4.2 Recommended logical business key

The logical business key for active-episode uniqueness is:

LOGICAL KEY Source organisation/system + receiving site + stable purchase-order identifier + stable purchase-order-line identifier + active-episode indicator. A generated episode sequence distinguishes linked historical recurrences. Risk type is not part of the uniqueness key because delay and stockout signals on the same PO line are normally one operational problem, not two cases.

A delivery-schedule identifier is retained as supporting risk-component evidence. It becomes part of a separate operational key only if the source treats schedules as independently actionable lines with different owners or sites; that is outside the frozen MVP assumption.

4.3 Treatment rules

Situation	Frozen treatment
PO amendments	Quantity/date/price or promise changes on the same stable source line update facts and append an amendment event within the active episode. They never overwrite the opening snapshot. Cancellation can support Demand Cancelled resolution. A technically deleted/recreated line with a new stable line ID creates a linked successor only when it represents a new obligation.
Split delivery schedules	All schedules for one PO line are assessed as risk components inside one active episode. The episode shows earliest threatened need date, residual open quantity by schedule and total exposure. It does not create one exception per schedule in the MVP.
Partial receipts	Apply receipts to the relevant schedule where a valid reference exists; otherwise apply by approved chronological allocation (earliest due open schedule first). Recalculate risk only on residual open quantity. A partial receipt is an event and may reduce severity; it is not resolution unless the protected requirement is fully covered under the selected resolution category.
Multiple receipt events	Append each receipt and allocation independently; retain source transaction, time, quantity, UOM and run. Aggregate only for current exposure. Never replace prior receipt history with a cumulative number.
Corrected/reversed receipts	Append a correction/reversal linked to the original receipt and recompute residual quantity. If not Closed, create a Reopened/Resolution Withdrawn event and return to Investigating. If Closed, a material recurrence creates a linked successor episode.
Repeated pipeline scoring	Create one candidate evaluation per eligible PO line/run. If an active episode exists, link the evaluation and update the current risk view without creating a new exception. Non-qualifying scores do not auto-resolve.
Closed risk later deteriorates	If a new candidate meets the material-recurrence rule after Closed, create a new episode sequence linked to the predecessor. Prior closure, evidence and SLA remain unchanged.
Suppressed risk	Candidate evaluations continue and link to the suppressed episode. No new exception opens while the suppression is valid. Material early-reactivation threshold or expiry returns the same pre-closure episode to active work.
PO line moved to another site	Treat as a new operational obligation and successor episode because site ownership, inventory exposure and SLA accountability change. Link both records.

4.4 Material recurrence and duplicate prevention

- **Active duplicate check:** Before opening, search for Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved or Suppressed episode with the logical key. If found, link the candidate; do not create.
- **Concurrent-run control:** Only one successful publication may claim the active key. A losing concurrent attempt links to the existing episode or fails safely; it cannot create a second case.
- **Material recurrence after closure:** A successor opens when the current qualifying severity is High/Critical, or the score increases by at least 15 points above the closing snapshot and crosses the opening threshold, or an objective adverse event reverses the closure basis. These are proposed portfolio thresholds.
- **Same-episode reopening:** Before closure, expiry, revocation, failed action or reversed evidence returns the existing episode to active work. No new episode is needed.
- **Post-closure reopening:** After closure, the Reopened event links predecessor and successor episode IDs. This provides one operational family without corrupting per-episode cycle time.
- **Duplicate reporting:** Candidate dispositions count linked-existing and suppressed-blocked evaluations separately from new exceptions; operational backlog counts the episode once.

5. Ownership and assignment model

5.1 Automatic assignment hierarchy

Priority	Mapping route	Rule
1	Approved supplier-site exception mapping	Use where a supplier relationship is deliberately centralised for a site and mapping is effective/active.
2	Buyer-group + site mapping	Primary default. Assign to the eligible procurement analyst responsible for the PO buyer group at the receiving site.
3	Product-category + site fallback	Use when buyer group is missing/unmapped but category stewardship is defined.
4	Site procurement duty queue	Use only when a staffed rotating duty owner is configured and acceptance is attributable.
5	Unassigned queue	If no eligible individual can be resolved, remain Open, display visibly and escalate; never assign to a fictitious generic owner.

Temporary delegation is applied after the base owner is resolved. It redirects new and active assignments only for the approved effective interval while retaining the base owner, delegate, approver and dates.

5.2 Ownership controls

Control area	Frozen policy
Buyer-group ownership	Accountable for supplier contact, plan completeness and case progression. It is the default operating-owner dimension.
Site ownership	Defines managerial accountability, calendar context and inventory-planning partner; it is not a substitute for an individual owner.
Supplier ownership	Permitted as an approved site-specific exception when central relationship ownership is more appropriate than buyer group.
Product-category ownership	Fallback only; never overrides a valid supplier-site or buyer-group mapping without an approved configuration change.
Unassigned queue	Open state, visible to site manager. Critical: manager event at 30 minutes and breach at 1 business hour. High: reminder at 2 hours and breach at 4 business hours.
Manual reassignment	Actor selects new eligible owner and reason. Current lifecycle stage is retained. Critical reassignment requires manager authorisation; reassignment never erases the prior owner or resets SLA.

Control area	Frozen policy
Temporary delegation	Requires delegate acceptance, manager approval, start/end, scope and reason. Expiry returns routing to the base owner but does not silently reassign cases already in progress unless specified in the approval.
Accountability after escalation	Current owner remains Responsible. Site supply-chain manager becomes Accountable for intervention until breach clears or the owner is explicitly changed.

5.3 Reassignment reason codes

- Workload balancing
- Temporary absence/delegation
- Incorrect mapping
- Supplier relationship ownership
- Site transfer
- Product-category expertise
- Conflict of interest
- Manager intervention
- Owner inactive
- Data correction

5.4 RACI matrix

Activity	Procurement analyst	Inventory planner	Supply-chain manager	Data/system admin	Governance/audit
Release validated source run	I	C	A	R	C
Maintain assignment mappings	C	C	A	R	C
Triage and investigate PO-line exception	R	C	A	I	I
Assess projected inventory exposure	C	R	A	I	I
Agree supplier/operational action plan	R	C	A	I	I
Approve material override/suppression	C	C	A/R	I	C; concurrence where required
Monitor commitments and evidence	R	R for stock actions	A	I	I
Resolve within category authority	R	R for inventory actions	A for controlled routes	I	I
Close normal exception	C	C	A/R	I	I
Approve exceptional closure	C	C	R	I	A/concurrence
Correct source/data processing defect	C	C	A	R	C
Review audit completeness/control design	I	I	C	C	A/R

R = Responsible; A = Accountable; C = Consulted; I = Informed. In the MVP these are simulated role identities and do not prove enterprise authentication or segregation of duties.

6. SLA and escalation policy

PROPOSED ASSUMPTIONS Every rule in this section is a portfolio modelling assumption, not an observed or approved company policy. It must be labelled as such in requirements, tests and demonstrations.

6.1 Business calendar

Assumption	Frozen definition
Time zone	Australia/Melbourne using the IANA time-zone rules; store/display all operational deadlines consistently and test daylight-saving transitions.
Business days	Monday-Friday excluding Victorian public holidays. The MVP uses one common control calendar; a future design may add site-specific overlays.
Operating hours	08:30-17:00 local time, 8.5 business hours per day.
Daylight saving	Use local wall-clock business intervals. The 08:30-17:00 day remains 8.5 business hours across DST changes; UTC offset is derived, never hard-coded.
After-hours creation	Record actual Open timestamp immediately. SLA consumption begins at the next business opening; after-hours critical events remain visible and may generate a non-SLA alert in later phases.
Public holidays	No SLA minutes accrue. If a holiday definition changes, future deadlines recalculate only under controlled policy; historical SLA results retain the calendar version used.

6.2 SLA clocks and targets

Clock	Starts	Stops	Target	Interpretation
Critical assignment	Open	Named owner recorded	1 business hour	Tighter than BRS common 4-hour target; still supports >=95% High/Critical within 4 hours.
High assignment	Open	Named owner recorded	4 business hours	Matches BRS assignment target.
Medium assignment/review	Open	Named owner and investigation begun	1 business day (8.5 hours)	Review expectation; not a crisis response.
Critical action plan	Open	First compliant Action Agreed	4 business hours	Includes assignment and investigation time.
High action plan	Open	First compliant Action Agreed	8 business hours	Includes assignment and investigation time.
Medium action plan	Open	First compliant Action Agreed or documented no-action resolution	2 business days (17 hours)	Proposed planning expectation.
Closure review	Resolved	Closed or resolution withdrawn	1 business day	Control review expectation; reported separately from response SLA.

6.3 Pause, reset and reassignment rules

Rule	Frozen treatment
Permitted pause	Critical source-data integrity hold or system unavailability that prevents safe investigation, approved by manager/admin with start, reason and expected end; or effective approved Suppressed state.
Not a pause	Waiting for supplier, planner workload, owner absence, approval delay, reassignment, weekends/holidays (already excluded), or missing internal follow-up.
Pause end	Automatically at suppression expiry/system recovery, or human release of data hold. Preserve consumed and paused minutes.
Reset	No reset within the same episode. A genuinely new linked successor after Closed starts new clocks. A prior breach remains a historical breach.
Severity increase	Apply the earlier of the existing due time or the new severity target measured from increase time; never extend a due time.

Rule	Frozen treatment
Severity decrease	Does not extend an existing due time unless a separately approved SLA extension exists. Original severity and due time remain reportable.
Reassignment	No reset or pause. New owner receives the remaining time; manager is accountable for critical/overdue reassignment.
Same-episode reopening	Resume remaining clock or retain breached condition. Do not reset.
Post-closure successor	Starts new clocks at successor Open because it is a new episode, while predecessor SLA remains frozen.

6.4 Reminder and escalation events

Control	Reminder	Breach/event	Escalation	Accountability
Critical assignment	30 min	1 business hour	Immediate manager queue; repeat every 1 business hour until owned	Owner once assigned; manager accountable during breach
Critical action plan	2h and 3h	4h	Manager intervention event; daily unresolved critical review	Owner remains responsible; manager accountable
High assignment	2h	4h	Manager queue; repeat at next business opening if still unassigned	Site manager accountable
High action plan	4h and 6h	8h	Manager intervention event	Owner remains responsible
Medium review	4h before due	8.5h	Manager queue next business opening	Owner responsible
Monitoring commitment	At 75% of interval and 1 business hour before due	At commitment/next-action due	Return to investigation decision required; critical breach also manager event	Owner and manager for critical
Suppression expiry	1 business day and 1 hour before expiry	At expiry	Automatic reactivation; manager event if Critical	Prior owner if eligible; manager if unassigned

7. Approval matrix

Approval changes authority, not evidence. Every request and decision retains requester, approver, time, reason, original/requested value, evidence, conditions and expiry where applicable. A rejection applies no change.

Action	When approval is required	Primary approver	Additional approval	Self-approval	Required basis	Constraints
Risk-priority reduction	Any override that lowers displayed operational priority or response obligation; calculated score stays unchanged	Supply-chain manager	Governance concurrence if Critical becomes Medium/Monitor	Yes	Reason, current score, requested band, mitigation evidence, expiry	Time-bound; cannot erase original SLA
Suppression	All manual active-episode suppression	Supply-chain manager	Governance concurrence for Critical >1 business day, repeated suppression, or second extension	Yes	Reason code, narrative, evidence, start/expiry, reactivation criteria	Maximum initial term: 5 business days; Critical: 1 business day
Product-criticality override	Any change to site/item criticality; downward change is material	Supply-chain manager	Governance concurrence for downward Critical change	Yes	Original/requested value, site scope, reason, evidence, effective period	Does not rewrite historical scores
Residual-risk acceptance	Resolution while material risk remains	Supply-chain manager	Governance concurrence for Critical or safety/production-critical impact	Yes	Residual exposure, consequence, controls, owner, expiry/review date	Acceptance expires; recurrence rule remains active
Closure without normal evidence	Missing expected objective evidence or broken reference	Supply-chain manager	Governance/audit reviewer mandatory	Yes	Why evidence unavailable, alternative corroboration, limitation and follow-up	Exceptional flag; cannot be used for convenience
Critical exception reassignment	Change accountable owner after Critical assignment	Supply-chain manager	Second manager/governance if manager requested the change	Requester may not be sole approver	Old/new owner, reason, capacity/skill basis, handover	No SLA reset
SLA extension	Change future operational due time beyond frozen target	Supply-chain manager	Governance concurrence if Critical or repeated	Yes	Cause outside reasonable owner control, revised due, control plan	Prospective only; no retrospective removal of breach; maximum one extension
Due-date extension	Change accepted supplier/demand date when it reduces risk or delays action	Supply-chain manager	Demand owner/planner consultation; governance for Critical	Yes	Authoritative change evidence, old/new date, operational impact	Original date retained; source correction distinguished from operational concession
Reject system recommendation	Reject deterministic suggestion or future AI output	Current authorised owner	None	No	Reason code and optional note	Rejection is not an override unless a controlled value/action changes
Future AI recommendation acceptance	Accept text/options for operational consideration	Authorised owner for relevant action	Any action-specific approval still applies	No self-approval of downstream controlled action	AI output reference, edits, reviewer decision	Acceptance alone cannot mutate state, priority, owner, suppression or execution

7.1 Approval status model

Status	Meaning	Lifecycle effect
Requested	Complete request awaits decision.	No requested change is effective.
Approved	Authority granted within stated scope/period.	Change may be executed through its separate controlled action.
Approved with conditions	Authority depends on documented conditions.	Execution blocked until conditions are satisfied or explicitly tracked.
Rejected	Authority denied with rationale.	No change; request and rejection remain auditable.
Expired	Approval validity ended.	Override/suppression ceases or returns for review; no silent extension.
Withdrawn	Requester withdrew before decision/execution.	No change; withdrawal event retained.

8. Resolution and closure evidence

RESOLUTION CONTROL All ten categories are structured outcomes, but they are not equally strong. Supplier Commitment Accepted is a monitored mitigation and may support Resolved only when the revised commitment objectively removes the current exposure and manager accepts any residual risk. It is not equivalent to receipt.

Resolution category	Required evidence	Authorised resolver	Approval	Monitoring period	Closure condition	Reopen trigger
Delivery received	Posted receipt transaction(s), quantity/UOM, time, PO/schedule link; residual quantity/exposure	Procurement analyst	None normally; manager if tolerance/over-receipt exception	One successful validated run after posting	Required protected quantity covered; no unresolved reversal; evidence verified	Receipt reversed/corrected; residual need exposed; new stockout signal
Supplier commitment accepted	Supplier commitment date/quantity, contact reference, reliability context, revised exposure	Procurement analyst	Manager if risk remains High/Critical or date concession reduces priority	Until next checkpoint and at least 1 business day; continue to delivery where Critical	Commitment places receipt before need with acceptable buffer; residual risk documented; for Critical, closure normally waits for receipt or formal acceptance	Commitment missed/withdrawn; reliability deteriorates; projected stockout returns
Alternative supplier arranged	Replacement order/commitment reference, approved supplier, quantity/date, cancellation/containment of original exposure	Procurement analyst	Manager; procurement authority outside system remains out of scope	Until alternative commitment is confirmed and one successful risk run	Alternative quantity/date covers exposure; approvals/evidence valid	Alternative supplier fails, date slips, or coverage becomes insufficient
Substitute product approved	Planner assessment, substitute item, equivalence/usage approval, quantity, reservation/availability	Inventory planner	Manager or designated product authority; Critical requires manager	One successful inventory/demand run after reservation	Approved substitute covers requirement through need date	Approval withdrawn; substitute inventory consumed; conversion/UOM defect
Inventory reallocated	Transfer/allocation/reservation reference, from/to site, quantity, availability and impact on donor demand	Inventory planner	Manager if donor site risk increases or Critical	One successful run after allocation/transfer confirmation	Receiving need covered without creating unaccepted donor risk	Transfer cancelled/late; donor risk becomes material; allocation released
Demand date changed	Authorised demand/production/customer change reference, old/new need date, impact assessment	Inventory planner or procurement analyst	Manager; demand owner consultation	One successful run using authoritative new date	New date is authorised and removes current exposure; original date retained	Date reverts/advances; change unapproved; risk crosses threshold

Resolution category	Required evidence	Authorised resolver	Approval	Monitoring period	Closure condition	Reopen trigger
Demand cancelled	Authoritative cancellation reference, affected quantity/site, source status	Procurement analyst or planner	Manager for Critical or partial cancellation ambiguity	One successful run confirming cancelled requirement	No remaining protected demand for affected quantity	Cancellation reversed; residual demand identified
Residual risk formally accepted	Residual exposure, consequence, controls, accountable owner, review/expiry, alternatives considered	Supply-chain manager	Governance concurrence for Critical/safety/production-critical	At least 1 business day unless immediate decision deadline documented	Approval valid; conditions met; expiry/review scheduled	Acceptance expires/conditions fail; consequence or exposure increases
False positive	Rule/fact analysis showing candidate validly scored but did not represent actionable exposure; rule-review reference	Procurement analyst	Manager for High/Critical; governance consulted for repeated pattern	One successful rerun or independent check	No actionable exposure and explanation retained; not merely low score	Same signal recurs materially; analysis disproved; rule defect confirmed
Source-data correction	Admin correction event, before/after values, source authority, defect code, corrected run	Procurement analyst confirms; admin performs correction	Manager for High/Critical closure; governance for exceptional evidence	One successful validated reprocessing run	Corrected authoritative data shows risk did not exist or has been removed	Correction reversed/invalid; new authoritative data recreates exposure

8.1 Closure verification checklist

- Exception is in Resolved; no direct active-to-Closed route.
- Resolution category, cause, resolver and resolution time are present.
- Evidence reference is resolvable and matches the affected PO line, site, quantity and dates.
- Category-specific monitoring period/condition is complete, or exceptional-closure approvals are attached.
- Residual risk is zero/acceptable or formally approved with review/expiry.
- No mandatory action, overdue monitoring checkpoint or unresolved receipt correction remains.
- Closure reviewer is authorised and distinct where the approval matrix requires separation.
- Final outcome, SLA result and limitations are captured; closure never rewrites prior events.

9. Five end-to-end scenarios

A. Overdue critical order resolved through receipt

Element	Walkthrough
Starting facts	Site A production-critical SKU; PO line 450012/20 has 100 units open, latest accepted promise was yesterday, projected stock reaches zero in two days; current score 82/Critical.
Automated events	Validated run creates candidate and Open episode; buyer-group/site mapping assigns Buyer A; assignment event at Open+5 minutes. Critical reminders/due times are scheduled. Later, two receipts of 60 and 40 units are posted; system appends both and flags possible resolution evidence after the second validated run.
Human actions and states	Open -> Assigned. Buyer A starts Investigating, records supplier contact and cause. Investigating -> Action Agreed with expedite plan and commitment; Action Agreed -> Monitoring when shipment is confirmed. Buyer reviews two receipts, verifies residual quantity zero and moves Monitoring -> Resolved using Delivery Received. Manager verifies one successful run and closes Resolved -> Closed.
Ownership	Buyer A remains Responsible; Site A supply-chain manager Accountable because Critical. No reassignment.
SLA	Assigned within 1 hour and Action Agreed within 4 business hours. Receipt monitoring deadline is separate. SLA stops at Action Agreed; final SLA result retained at closure.
Approvals	No material override. Manager acts as closure reviewer; no exceptional evidence route.
Audit events	Run accepted; candidate scored; exception opened; assigned; investigation started; supplier contact; plan agreed; monitoring started; receipt 1; partial exposure recalculated; receipt 2; possible-resolution evidence; Resolved; closure review; Closed.
Final outcome	Closed with Delivery Received, 100 units received, no residual exposure. Any later receipt reversal creates a linked successor episode because formal closure occurred.

B. Stockout risk mitigated through product substitution

Element	Walkthrough
Starting facts	Site B customer-critical SKU has 50 units due in three days; projected balance is -30 before expected PO receipt; score 74/Critical. Approved substitute exists but is not yet reserved.
Automated events	Candidate opens one episode and assigns the buyer-group owner. System attaches stock projection and substitute availability as evidence but makes no decision.
Human actions and states	Buyer starts Investigating and requests planner assessment. Planner confirms equivalence and 40 substitute units; buyer confirms the remaining 10 from existing inventory. Investigating -> Action Agreed after manager/product authority approval. Action Agreed -> Monitoring when reservation is posted. After one validated inventory run confirms coverage, planner moves Monitoring -> Resolved using Substitute Product Approved. Manager closes after evidence review.
Ownership	Procurement analyst remains case owner; inventory planner is Responsible for the substitution assessment; manager Accountable.
SLA	Critical assignment and four-hour action-plan targets apply. Planner consultation does not pause SLA.
Approvals	Substitute approval by designated product authority/manager. Any criticality or priority reduction would require a separate request; none is implied by substitution.
Audit events	Candidate/open/assignment; investigation; planner assessment; substitution approval request/decision; action agreed; reservation; monitoring; validated coverage; Resolved; Closed.
Final outcome	Closed with substitute coverage of all 50 units. Release/consumption of reserved stock later triggers material recurrence and a linked successor if already closed.

C. Supplier commitment followed by monitoring and later breach

Element	Walkthrough
Starting facts	High exception: 200 units are at risk; supplier promises delivery in five business days, one day before need date.
Automated events	Run creates and assigns the episode. Monitoring reminders are scheduled. At the commitment date, no receipt/ASN confirmation exists; system records monitoring breach and manager escalation but does not change the lifecycle state.
Human actions and states	Open -> Assigned -> Investigating. Buyer records the promise and selects Action Agreed, then Monitoring after commitment acknowledgement. On breach, owner moves Monitoring -> Investigating, records supplier failure and agrees alternate supplier action. The episode remains the same; no duplicate is created.
Ownership	Original buyer remains Responsible. Manager becomes accountable at breach. If reassigned, reason and handover are recorded and SLA does not reset.
SLA	High Action Agreed within 8 business hours. Waiting on supplier never pauses SLA or monitoring. Breach event remains visible after recovery.
Approvals	Alternative supplier arrangement requires manager approval in this portfolio; any due-date concession that reduces urgency is a separate approved action.
Audit events	Promise recorded; action plan; monitoring; reminders; commitment breach; escalation; return to Investigating; new plan/approval; subsequent evidence and eventual resolution/closure if successful.
Final outcome	The initial commitment does not close the case. It returns to active investigation on breach, demonstrating that escalation and breach are events rather than statuses.

D. Exception suppressed with an expiry date and later reopened

Element	Walkthrough
Starting facts	Medium exception arises from a planned short supplier shutdown already covered by temporary safety stock. The owner requests three business days of suppression to avoid duplicate daily work.
Automated events	Candidate creates Open and assignment. System continues daily candidate evaluations while suppression is active, links them to the same episode and sends expiry reminders. At expiry it reactivates automatically.
Human actions and states	Open -> Assigned -> Investigating. Owner submits suppression with evidence and expiry; distinct manager approves. Investigating -> Suppressed. At expiry, the owner mapping is still valid, so system records Reopened/Suppression Expired and moves Suppressed -> Assigned. Owner begins Investigating and either resolves with evidence or creates a plan.
Ownership	Base owner is preserved during suppression; manager remains approval/accountability owner. If the base owner becomes inactive, expiry returns to Open/unassigned.
SLA	Consumed time is preserved. Effective suppression pauses the action clock; expiry resumes remaining time. It does not reset, and a prior breach remains breached.
Approvals	Manager approval, no self-approval, mandatory reason/evidence/expiry. Extension requires a new request; repeated or Critical suppression needs governance concurrence.
Audit events	Suppression requested; approved; state changed; daily linked candidates; expiry reminders; expiry; reactivation/Reopened event; Assigned; Investigation.
Final outcome	Same pre-closure episode resumes. No new duplicate is created and no suppressed history is deleted.

E. Source-data correction invalidates the original exception

Element	Walkthrough
Starting facts	A High overdue exception opened because an accepted promise date was loaded as 10 July instead of 10 August. The source extract is demonstrably invalid.
Automated events	Initial run validly scores the supplied data and creates the episode. After admin correction, a new run passes validation, appends corrected facts and flags possible Source-Data Correction resolution evidence. It does not auto-resolve.
Human actions and states	Owner starts Investigating, identifies date inconsistency and links a data-defect record. Administrator appends correction with before/after and source authority. Owner reviews corrected run and moves Investigating -> Resolved using Source-Data Correction. Manager reviews High closure evidence and moves Resolved -> Closed.
Ownership	Procurement analyst remains operational owner; administrator owns correction; manager is accountable for closure. Governance may review repeated data defects.
SLA	SLA may pause only after manager/admin approve a critical data-integrity hold. Time before approved hold remains consumed. Correction/reprocessing time outside the hold is counted.
Approvals	Normal correction is attributable, not a priority override. High closure requires manager review; closure without authoritative correction evidence would require exceptional route.
Audit events	Original source/run/candidate/open; assignment; investigation; defect raised; approved hold if used; admin before/after correction; corrected run; possible-resolution evidence; Resolved; Closed.
Final outcome	Closed as Source-Data Correction, not silently deleted as a false alert. Original score remains explainable against the originally supplied values.

10. MVP simplification review

Recommendation	Scope	Critical assessment
Must remain in MVP	Eight-state controlled lifecycle; candidate-to-episode link; one active episode per PO line; assignment/unassigned queue; human Resolved/Closed; suppression reason/expiry/approval; append-only event history; reason/evidence references; negative transitions; proposed SLA clock and breach events; role simulation labels.	These establish the governance story and central portfolio value.
May be simulated	Role switching; holiday calendar fixtures; temporary delegation; two-person approvals using distinct demo roles; evidence as synthetic metadata/reference rather than document repository; manager escalation as in-app event/queue; receipt reversals and concurrent-run conflicts as test fixtures.	The portfolio can demonstrate semantics without enterprise infrastructure.
Should be deferred	Multi-level approval engine, jurisdiction-specific calendars for both sites, configurable business-rule designer, supplier portal, email/Teams, live ERP integration, sophisticated schedule-line case management, WORM audit storage, advanced identity/RBAC, hosted multi-user operation, AI recommendations.	They expand technology and control claims without strengthening the first vertical slice.
Must not be removed	Stable episode identity; no active duplicates; distinction between candidate/exception; human resolution and closure; expiry on suppression; no self-approval for material overrides; original score/history retention; corrected/reversed receipt handling; no SLA reset on reassignment; evidence and actor/timestamp/before-after/reason on controlled events.	Removing these would make the case-management and internal-control claims materially weaker.

10.1 Practical MVP implementation boundary

- Use one common Melbourne business calendar and a small, frozen set of SLA calculations rather than a general calendar engine.

- Implement only the listed transitions and ten resolution categories; do not build a configurable workflow designer.
- Represent approvals as one structured request/decision record, with a second concurrence only for the few explicitly identified Critical cases.
- Use synthetic evidence references and validate their metadata; do not build document storage.
- Demonstrate one current owner plus delegation/reassignment history; do not implement workload optimisation.
- Test five golden scenarios plus negative controls and reversal/duplicate boundaries before any notification, deployment or AI feature.

10.2 Minimum governance evidence for the MVP

Evidence pack	Minimum demonstration
Golden lifecycle	At least one case traverses Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved and Closed with complete event fields.
Suppression/reopening	One approved suppression expires and returns the same pre-closure episode to active work without SLA reset.
Linked recurrence	One Closed episode remains Closed while a material recurrence opens exactly one linked successor.
Duplicate control	Repeated and concurrent candidate fixtures result in one active episode and multiple linked evaluations.
Receipt integrity	Partial, multiple and reversed receipt fixtures produce correct residual exposure and reopening behavior.
Negative authority	Open/Assigned-to-Closed, self-approval, expiry-free suppression, automated Resolved/Closed and history deletion attempts all fail with zero state mutation.
SLA boundaries	After-hours, holiday, DST, pause, reassignment and reopening fixtures reproduce the frozen business-minute results.
Evidence limitation	Screens and portfolio documentation label identities, approvals, SLAs and outcomes as simulated controls, not production validation.

11. Frozen decision register

Decision ID	Decision	Rationale	Alternatives rejected	Requirements effect	Later modelling effect	Remaining uncertainty
LIF-DEC-001	Use eight operational states: Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved, Closed, Suppressed.	Clear work stages with temporary dormancy.	Candidate/Reopened/Escalated as states.	Revise LIF-010/matrix.	State history must separate events/conditions.	Practitioner usability not validated.
LIF-DEC-002	Candidate Risk is analytical, never a lifecycle status.	Preserves scored-population traceability without inflating backlog.	Candidate as New status.	Clarify LIF-001/002 and queue definitions.	Candidate-to-episode relationship required.	Opening thresholds remain rule-catalogue decision.
LIF-DEC-003	Resolved and Closed are human-controlled in MVP.	Corrections/reversals make automatic resolution unsafe; user baseline is binding.	Auto-resolve full receipt.	Revise LIF-007 and acceptance tests.	Possible-evidence event distinct from transition.	Future automation could be reconsidered only by change control.
LIF-DEC-004	Closed is immutable completion; post-closure recurrence creates linked successor.	Protects outcome/cycle-time evidence.	Change same closed episode to Reopened state.	Revise FR-011/023 and LIF-010.	Episode sequence/predecessor relationship needed.	Material recurrence threshold is proposed.
LIF-DEC-005	Suppressed is a temporary active-episode dormant state with mandatory expiry.	Noise control without hidden indefinite risk.	Delete/close/cooldown as status.	Strengthen LIF-009.	Suppression control, prior state and SLA consumption required.	Maximum durations are assumptions.
EPI-DEC-001	One episode covers materially related delay and stockout exposure on one stable PO line/site until Closed.	Avoids parallel cases for one operational problem.	Separate delay and stockout exceptions.	Refine FR-011/RULE-013.	Logical active key and risk components required.	Schedule-line exception may be needed in real ERP.
EPI-DEC-002	Split schedules are components inside one PO-line episode in MVP.	Matches confirmed operational grain and solo scope.	One episode per schedule.	Clarify RULE-014.	Schedule evidence/allocations retained but not active uniqueness.	Depends on schedules not having separate owners/sites.
DUP-DEC-001	One active episode per logical PO-line/site key; repeated scores link to it.	Prevents workload/reporting duplicates.	New exception per run or risk signal.	Strengthen FR-012.	Active uniqueness and candidate disposition required.	Concurrency proof is simulated.
RCV-DEC-001	Risk uses residual open quantity; partial receipts are append-only events allocated to schedules.	Operational exposure is quantity-specific.	Treat any receipt as resolved; overwrite cumulative receipt.	Refine DQ-003/RULE-014/FR-021.	Receipt event/allocation and residual calculation required.	Allocation tolerance remains source-policy assumption.
RCV-DEC-002	Receipt corrections/reversals reactivate before closure or create successor after closure.	Objective evidence is not irrevocable.	Ignore correction after closure.	Add lifecycle tests and BRS language.	Correction linkage and recurrence trigger required.	Reversal latency window not empirically validated.
OWN-DEC-001	Assignment precedence: supplier-site exception, buyer-group/site, category/site fallback, duty queue, unassigned.	Deterministic hierarchy consistent with BRS default and exceptions.	Pure manual or supplier-only ownership.	Expand SLA-001.	Effective-dated mappings and trace required.	Real organisation may prefer PO buyer first.
OWN-DEC-002	Unresolved mappings remain Open/unassigned and escalate visibly.	Generic ownership hides control gaps.	Assign to system/admin/default user.	Clarify LIF-003/SLA-006.	Unassigned condition and manager accountability required.	Coverage target depends on synthetic mappings.
OWN-DEC-003	Reassignment retains lifecycle state and never resets SLA.	Prevents gaming and duplicate work.	Return to Assigned and restart clocks.	Freeze SLA-007.	Owner history plus current owner required.	Critical approval overhead may be simplified in MVP.
OWN-DEC-004	Temporary delegation is effective-dated and manager-approved; base owner retained.	Supports absences without losing accountability.	Unrecorded substitution.	Add ownership requirement.	Delegation scope/effective dates required.	May be simulated only.
SLA-DEC-001	Melbourne calendar: Mon-Fri 08:30-17:00, Victorian holidays, IANA DST.	Freezes BRS open calendar decision.	Elapsed time; hard-coded UTC offset.	Expand SLA-005/NFR-006.	Calendar version and business-minute logic required.	Two-site local holiday difference deferred.
SLA-DEC-002	Critical assignment 1h/action plan 4h; High 4h/8h; Medium review 1 day/action 2 days.	Differentiates urgency and preserves BRS targets.	Single 4h assignment target only.	Refine SLA-002/003/004.	Multiple due times and SLA results required.	Targets are portfolio assumptions.

Decision ID	Decision	Rationale	Alternatives rejected	Requirements effect	Later modelling effect	Remaining uncertainty
SLA-DEC-003	SLA starts at Open; after-hours consumption begins next business opening.	Opening is first operational obligation.	Start at assignment or investigation.	Clarify LIF-002/SLA-005.	Open event and calculated due times required.	No 24/7 critical coverage assumed.
SLA-DEC-004	Only approved data/system hold or effective suppression pauses; supplier waiting does not.	Avoids owner-controlled clock gaming.	Pause for all external waiting.	Expand SLA-005.	Pause intervals/reasons/approvals required.	Real company may use separate clocks.
SLA-DEC-005	No same-episode reset; successor after Closed starts new clock.	Protects performance evidence while separating recurrence.	Reset on reassignment, reopen or severity change.	Freeze SLA-007/LIF-010.	Clock history and predecessor link required.	Reopen performance reporting needs definition.
SLA-DEC-006	Escalation and breach are events/conditions; owner remains responsible, manager becomes accountable.	Stage and accountability remain visible.	Escalated lifecycle state or silent owner transfer.	Refine SLA-006.	Event/accountability link required.	Notification delivery deferred.
APR-DEC-001	Material approvals prohibit requester self-approval; Critical exceptional routes may require governance concurrence.	Separation protects risk-reducing decisions.	Manager self-approves own request.	Expand GOV-004/SLA-008.	Requester/approver/concurrence identities required.	Simulated roles cannot prove true SoD.
APR-DEC-002	Priority overrides do not overwrite calculated score and are time-bound.	Maintains explainability and comparability.	Edit score/band directly.	Strengthen FR-020/GOV-005.	Original/requested/effective/expiry required.	Operational display convention to be designed later.
APR-DEC-003	SLA extensions are prospective, once only, and never remove a recorded breach.	Prevents retroactive performance manipulation.	Edit due time or clear breach.	Add to SLA-008/009.	Original/revised due and approval required.	Allowed maximum needs real policy validation.
RES-DEC-001	Use ten structured resolution categories with category-specific evidence and reopen triggers.	Enables consistent outcomes and audit.	Free-text resolution only.	Expand FR-021 and tests.	Resolution category/evidence fields required.	Monitoring periods are assumptions.
RES-DEC-002	Supplier commitment is weaker than receipt and requires monitored evidence/residual-risk control.	A promise can fail and should not be treated as delivery.	Close immediately on promise.	Clarify resolution semantics.	Commitment/checkpoint/breach data required.	Supplier reliability threshold remains open.
RES-DEC-003	Closure only from Resolved after evidence review; exceptional closure requires manager + governance.	Formal completion is a control sign-off.	Active-to-Closed or score-based closure.	Strengthen FR-022/LIF-008.	Closure reviewer/evidence/exception flag required.	True independent review is simulated.
AUT-DEC-001	Automation may create/open/assign/flag/expire/reactivate but not resolve, close, suppress approval or accept risk.	Preserves human accountability.	Objective-event automatic resolution.	Revise LIF-007 and AI language alignment.	Actor type and proposed-evidence event required.	Future phase may revisit limited automation.
AUD-DEC-001	Every controlled transition/action records actor, Melbourne time, before/after, reason/source and evidence/approval link; corrections append.	Meets reconstructability objective.	Mutable current-state history only.	Confirm GOV-001/002/010.	Event lineage required.	Admin-level immutability cannot be proven.

12. Critical self-review and BRS revisions

12.1 Critical self-review

Review area	Finding and required treatment
Ambiguous transition	Investigating/Action Agreed may resolve without Monitoring for immediate objective outcomes. This is deliberate, but acceptance tests must prevent it becoming a shortcut around missing evidence.
Ambiguous suppression return	Returning Suppressed to Assigned rather than the exact prior stage loses stage continuity but avoids silently assuming prior investigation/action remains valid. The Reopened event retains the prior state for review.
Over-engineering risk	Ten approval actions, concurrence rules and multiple clocks could exceed a solo MVP. Implement one generic approval structure and only the explicitly required role checks; do not create a workflow engine.
Over-engineering risk	Temporary delegation, concurrent-run uniqueness and holiday calendars should be demonstrated through narrow fixtures rather than fully configurable administration.
Cannot genuinely prove	Simulated role switching cannot prove authentication, segregation of duties, identity lifecycle or resistance to administrator tampering.
Cannot genuinely prove	Synthetic evidence and scenario tests cannot validate whether managers would accept the thresholds, SLAs, resolution categories or monitoring periods in practice.
Cannot genuinely prove	Append-only UI behavior demonstrates a control pattern; it does not prove immutable/WORM audit, legal retention, backup recovery or database-admin separation.
Duplicate-work risk	Creating a new episode for each schedule, risk type, score run or post-closure change would fragment work. The logical active key and linked-successor policy are therefore binding.
Duplicate-work risk	If buyer-group, supplier and category mappings are applied independently instead of by precedence, multiple owners could believe they are responsible. One current owner is mandatory.
Automation/accountability risk	Auto-assignment can create false confidence if the mapped person is inactive; eligibility and unassigned fallback must be visible.
Automation/accountability risk	Objective receipt flags can be reversed or misallocated. Human Resolved/Closed preserves judgement but adds workload; this is an intentional MVP trade-off.
Automation/accountability risk	A risk score decrease may pressure users to resolve prematurely. Resolution requires evidence, not the current band.
Charter/BRS inconsistency	BRS LIF-007 permits objective automatic resolution, but the commissioning baseline requires human-controlled Resolved. This model supersedes that preliminary path and requires BRS revision.
Charter/BRS inconsistency	BRS presents Reopened as a state and FR-011 is ambiguous about same versus linked episodes. This model makes it an event and uses a successor only after Closed.
Charter/BRS inconsistency	BRS permits Candidate Risk -> Suppressed as a transition. Candidate disposition and operational Suppressed state must be distinguished to avoid creating fake suppressed cases.
Residual weakness	Supplier Commitment Accepted is a risky resolution category. Closure should normally wait for receipt for Critical cases; otherwise formal residual-risk acceptance is required.

12.2 BRS requirements requiring revision

BRS item	Required revision before technical modelling
FR-011	Replace ambiguous 'linked reopened episode/state' wording with: same episode reactivates before Closed; material recurrence after Closed creates a linked successor episode.

BRS item	Required revision before technical modelling
FR-012	Define logical active key at PO line/site level; risk type and split schedule are components, not separate active exceptions in MVP.
FR-019	Reference the frozen matrix; require rejected transition attempts to cause zero mutation and be logged where configured.
FR-020	Add SLA extension, due-date extension, critical reassignment, suppression, criticality and exceptional closure; preserve original values and expiry.
FR-021	State that Resolved is human-controlled in MVP; add category-specific evidence, monitoring and reopen triggers.
FR-022	Add closure-review checklist, normal/exceptional evidence routes and governance concurrence for exceptional closure.
FR-023	Change Reopened from a status concept to an event; define same-episode pre-closure reactivation and linked successor after closure.
DQ-003	Add receipt allocation, corrections/reversals, UOM/tolerance treatment and append-only event requirements.
GOV-002	Explicitly require previous state, new state, reason and evidence/approval reference on every transition, plus prior/new owner/value for same-state control events.
LIF-001	Clarify candidate dispositions and that candidate suppression is not an operational lifecycle state.
LIF-007	Remove automatic Monitoring -> Resolved. Replace with automated possible-resolution-evidence event and human transition.
LIF-009	Limit operational suppression to Open through Monitoring; require approval, expiry, evidence, prior-state retention and automatic reactivation. Separate candidate blocking/dedup disposition.
LIF-010	Rename to Reopening event policy and remove Reopened from state list.
Lifecycle matrix	Replace preliminary matrix with Section 3, including rapid-resolution paths, resolution withdrawal, suppression expiry and linked successor after Closed.
SLA-001	Insert frozen assignment precedence and eligible-owner/unassigned controls.
SLA-002	Differentiate proposed Critical 1-hour and High 4-hour assignment targets while retaining aggregate >=95% within 4 hours.
SLA-003/004	Add Medium review/action expectations and clarify that action-plan clock begins at Open.
SLA-005	Insert operating hours, Victorian holiday assumption, after-hours start, DST handling, pause rules and calendar version.
SLA-006	Clarify escalation is an event/accountability overlay, not status or silent ownership change.
SLA-007	Freeze no-reset rule for reassignment and same-episode reactivation; successor after Closed starts new clocks.
SLA-008/009	Expand approval scope and add prospective-only extension, no retrospective breach removal, concurrence and expiry rules.
RULE-014	Define residual quantity, split-schedule aggregation, chronological allocation fallback, multiple receipts and correction/reversal treatment.
Appendix C lifecycle row	Replace ten-state model reference with eight operational states plus analytical/event concepts.

12.3 Final frozen outputs

Required output	Location and status
A. Recommended final lifecycle	Eight operational states and non-state concepts are frozen in Section 2.
B. Complete state-transition matrix	All permitted transitions and negative controls are frozen in Section 3.

Required output	Location and status
C. Ownership and SLA policy	Assignment hierarchy, RACI, calendar, clocks, pause/reset and escalation rules are frozen in Sections 5-6.
D. Approval and resolution matrix	Authority, separation, evidence, monitoring, closure and reopen triggers are frozen in Sections 7-8.
E. Frozen decision register	Twenty-eight decisions with modelling effects and uncertainty are recorded in Section 11.
F. BRS revision list	Twenty-three targeted revisions are identified in Section 12.2.

RELEASE CONDITION This operating model is suitable as a semantic baseline for logical data modelling only after the project owner accepts the proposed assumptions. It is not production-ready, not validated by a real organisation and not evidence of realised business performance.

Source references

- Supply Chain Exception Control System - Project Charter, Version 1.0, 19 July 2026: business context, objectives, scope, limitations, success criteria and unresolved decisions D3-D12.
- Supply Chain Exception Control System - Business Requirements Specification, Version 1.0, 19 July 2026: FR-011 to FR-024, GOV-001 to GOV-010, LIF-001 to LIF-012, SLA-001 to SLA-009, RULE-013/014, Appendices C-E.